CPSC 481
Artificial Intelligence

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Group

Syllabus

Week 2 : Topics

- Deep dive into the Turing Test
- Rationality in AI
- Philosophical Foundations of AI
- Limitations of Early Symbolic Approaches

The Turing Test

- Proposed by Alan Turing in 1950
- Designed to sidestep the question "Can a machine think?"
- A computer passes if a human interrogator cannot tell if responses come from a person or computer

Capabilities Needed for Turing Test

- Natural language processing
- Knowledge representation
- Automated reasoning
- Machine learning

- For Total Turing Test:
  - Computer vision
  - Robotics

Why Not Pursue the Turing Test?

- Focus on underlying principles of intelligence rather than imitation
- Analogy to artificial flight: Success came from aerodynamics, not imitating birds

Thinking Rationally: Laws of Thought

- Aristotle codified "right thinking" with syllogisms
- Logic: Formal notation for statements about objects and relations
- Logicism: Build intelligent systems using logic programs
- Challenges: Uncertainty in real-world knowledge

Acting Rationally: Rational Agent Approach

- Agent: Something that acts
- Rational agent: Acts to achieve best outcome or best expected outcome under uncertainty
- Advantages: More general and amenable to scientific development than other approaches

Philosophical Foundations of AI

- Can formal rules draw valid conclusions? (Aristotle, logic)
- How does mind arise from physical brain? (Dualism vs. Materialism)
- Where does knowledge come from? (Empiricism, induction)
- How does knowledge lead to action? (Aristotle's deliberation)

Philosophical Questions in AI

- Weak AI vs. Strong AI: Task-specific vs. General intelligence
- Can machines really think? (Chinese Room argument)
- Ethics of AI: Utilitarianism vs. Deontological ethics
- Beneficial AI: Aligning machine objectives with human values

Limitations of Early Symbolic Approaches

- Lack of self-learning: Relied on hand-coded rules, no improvement over time
- Scalability issues: Struggled with complex, real-world problems
- Symbol grounding problem: Difficulty connecting symbols to real-world meanings
- Handling uncertainty: Boolean logic inadequate for ambiguous information

History: Dose of Reality (1966-1973)

- Early systems failed on difficult problems
- Reasons: Reliance on introspection rather than analysis; Intractability of combinatorial explosion
- Example: Perceptrons limited in representation

Transition to Modern Approaches

- Shift to probabilistic reasoning and machine learning
- From weak methods to domain-specific knowledge
- Emphasis on experimental results and real-world applications

Examples of Early Symbolic AI Systems

- SHRDLU: Natural language understanding in a blocks world
- Expert Systems: MYCIN for medical diagnosis
- General Problem Solver (GPS): Means-ends analysis

AI Winters

- First AI Winter (1974-1980): Funding cuts due to unmet expectations
- Second AI Winter (1987-1993): Collapse of Lisp machine market
- Lessons learned: Need for realistic goals and scalable methods

Philosophical Debates in AI

- Consciousness in Machines: Can AI be conscious?
- Free Will vs. Determinism: Implications for AI decision-making
- Mind-Body Problem: How computation relates to mental states

Discussion Questions

- What are the main limitations of the Turing Test as a measure of intelligence?
- How does the concept of rationality differ from human-like thinking in AI?
- Is philosophy relevant to building practical AI systems? Why or why not?

Group Activity

- Form groups of 4-5 students
- Discuss: Examples of symbolic AI failures and how modern AI addresses them
- Share one key insight with the class after 10 minutes

Detailed Look at Chinese Room Argument

- Thought experiment by John Searle (1980)
- Person in a room manipulates Chinese symbols without understanding
- Argues that syntax (rules) is not sufficient for semantics (meaning)
- Challenges strong AI claims

Responses to Chinese Room Argument

- Systems Reply: The whole system (person + room + rules) understands
- Robot Reply: Embodiment is needed for true understanding
- Brain Simulator Reply: Simulate neurons instead of symbols

Rationality in Practice

- Utility Theory: Maximizing expected utility
- Decision Theory: Making choices under uncertainty
- Examples: Poker-playing AI, stock trading algorithms

Limitations of Symbolic AI: Specific Problems

- Frame Problem: Determining what doesn't change in a situation
- Qualification Problem: Handling all possible exceptions to rules
- Ramification Problem: Dealing with side effects of actions

Shift to Sub-Symbolic AI

- Connectionism: Inspired by brain's neural structure
- Neural Networks: Learning from data rather than rules
- Emergence of Machine Learning in the 1990s

Introduction to AI Ethics

- From Chapter 27: The Ethics of AI
- Issues: Bias in AI, job displacement, autonomous weapons
- Principles: Transparency, accountability, fairness

Summary of Week 2

- Turing Test as a benchmark for human-like AI
- Rationality as a formal approach to intelligence
- Philosophical questions shaping AI development
- Limitations driving evolution from symbolic to modern AI

Preview of Week 3: Intelligent Agents

- Defining agents and environments
- PEAS framework
- Types of agents: reflex, model-based, goal-based, etc.

Homework Assignment

- Read Chapter 2: Intelligent Agents
- Write a short reflection on the Chinese Room argument (1 page)
- Due next class

Additional Resources

- Turing's original paper: "Computing Machinery and Intelligence"
- Searle's Chinese Room paper
- Online: AI Philosophy playlist on YouTube

References

- Russel and Norvig, Artificial Intelligence: A Modern Approach, 4th edition, Prentice Hall, 2021.
- Axiom Framework https://framework.axiomaticinfo.com